Result Sheet

Assignment: Standard Test
Student: Dhruv Tyagi
Total Score: 8.5/32

Question	Score	Reason
1	0/1	Incorrect option chosen. Student selected C (2.7 cm) instead of B (2.16 cm).
2	1/1	Correct answer B (4.4 cm).
3	1/1	Correct answer D (50°).
4	0/1	Incorrect option chosen. Student selected C (12:21) instead of B (4 : 7).
5	1/1	Correct answer D ($DC^2 = CF \cdot AC$).
6	1/1	Correct answer C (x = 16, y = 8).
7	0/1	Incorrect option chosen. Student selected A instead of D (Assertion is false as DE || BC, Reason is true).
8	1/1	Correct answer A (Both A and R are true and R is the correct explanation of A).
9	0.5/2	Identified common angle ∠B, but incorrectly stated ∠A = ∠A as common, failed to prove ΔABD ~ ΔCBA properly, and did not prove the required result $AB^2 = BC \times BD$.
10	1.5/3	Correctly proved ΔAOB ~ ΔCOD using AA similarity (1 mark). Set up the ratio of corresponding sides, but inverted the ratio when substituting (1/2 = CD/4 instead of 4/CD), leading to an incorrect result of DC = 2 cm instead of 8 cm (0.5 mark).
11	0/3	Did not perform the necessary construction (DG || BF) and wrote incomplete, incorrect similarity statements. No valid proof steps shown.
12	0/5	Only drew an incomplete construction diagram without drawing the diagonal BD or using the converse of BPT to establish parallel sides.
13	1/5	Rearranged the given relation $AD^2 = BD \times DC$ into ratios (1 mark), but erroneously assumed ∠B = ∠C = 45° without proof to arrive at ∠BAC = 90°.
14.i	0/1	Incorrect trigonometric setup and values; failed to find BC = 56.4 m.
14.ii	0/1	Used Pythagoras theorem with incorrect side lengths instead of tan 45° in right ΔAGF; did not find AG = 38 m.
14.iii	0/2	Not attempted.
14.iv	0.5/2	Identified the formula for total length (AF + FE + EB) for 0.5 marks, but calculated EB incorrectly using arbitrary values instead of trigonometry, yielding an incorrect final length.